MIL-M-26542/7A
29 March 1985
SUPERSEDING
MIL-M-26542/7
29 August 1983

MILITARY SPECIFICATION SHEET

MICROPHONE, M-948/A

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the microphone described herein shall consist of this specification and the latest issue of MIL-M-26542.

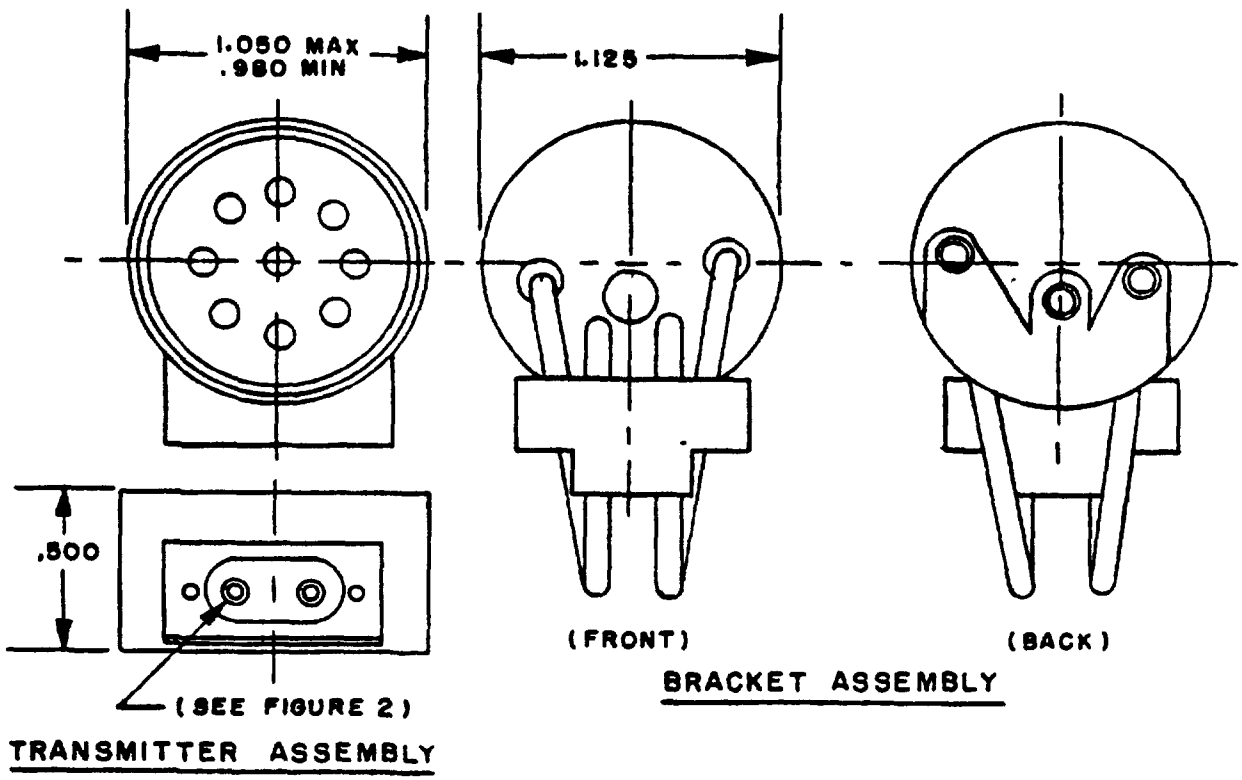
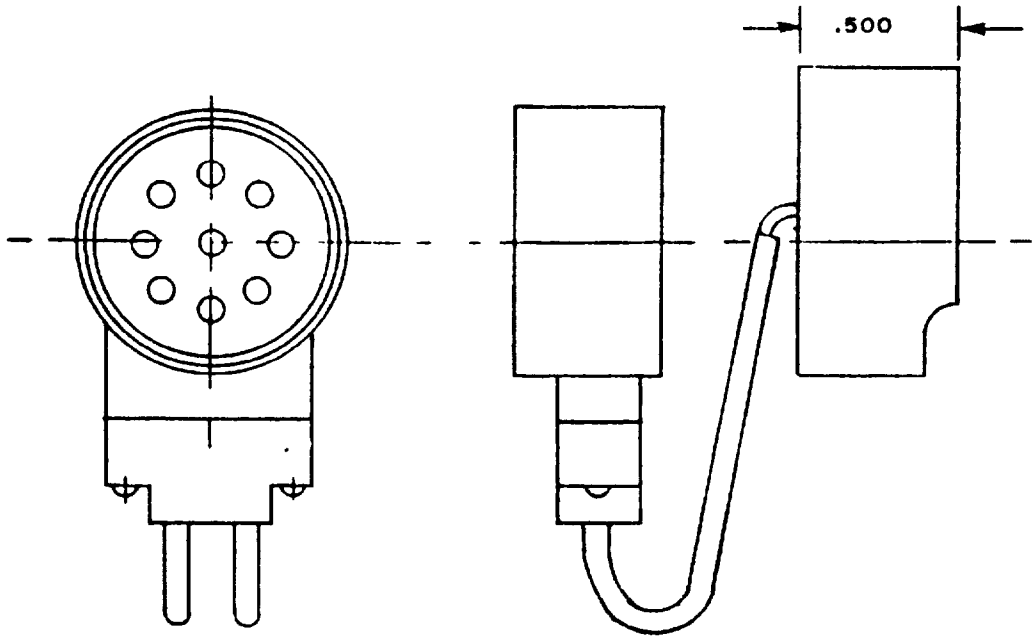


FIGURE 1. Microphone assembly.

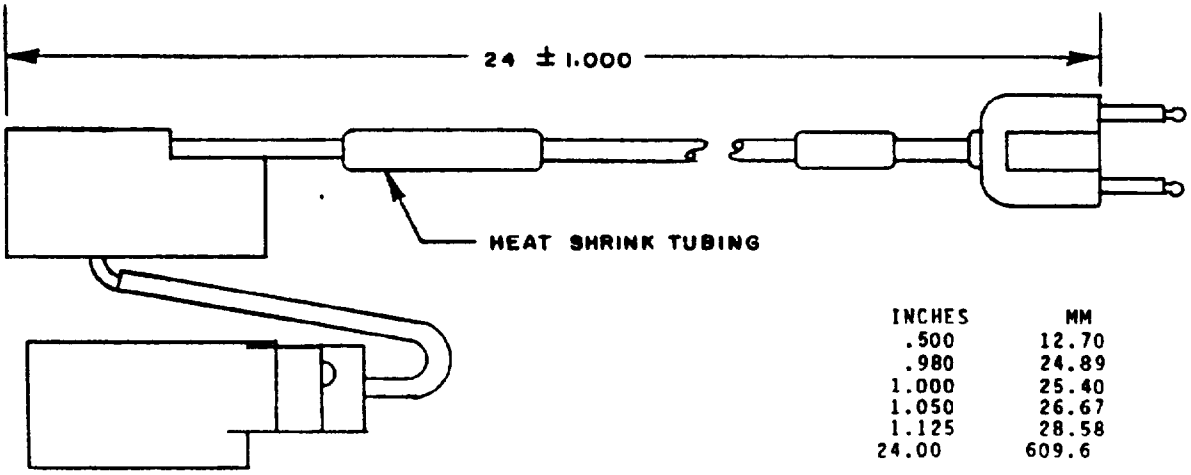
Ⓐ denotes changes

FSC 5965

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TRANSMITTER & BRACKET ASSEMBLY



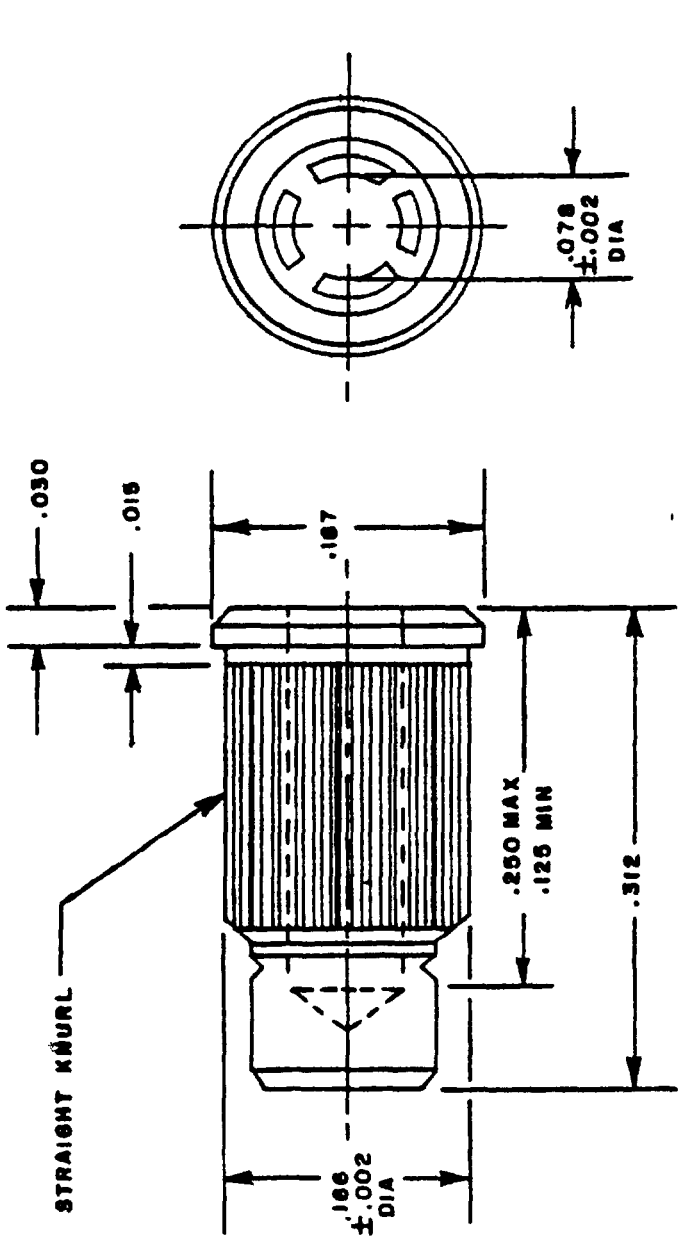
INCHES	MM
.500	12.70
.980	24.89
1.000	25.40
1.050	26.67
1.125	28.58
24.00	609.6

MICROPHONE ASSEMBLY
(SIDE VIEW)

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.015 (0.38 mm).

FIGURE 1 Microphone assembly - Continued.

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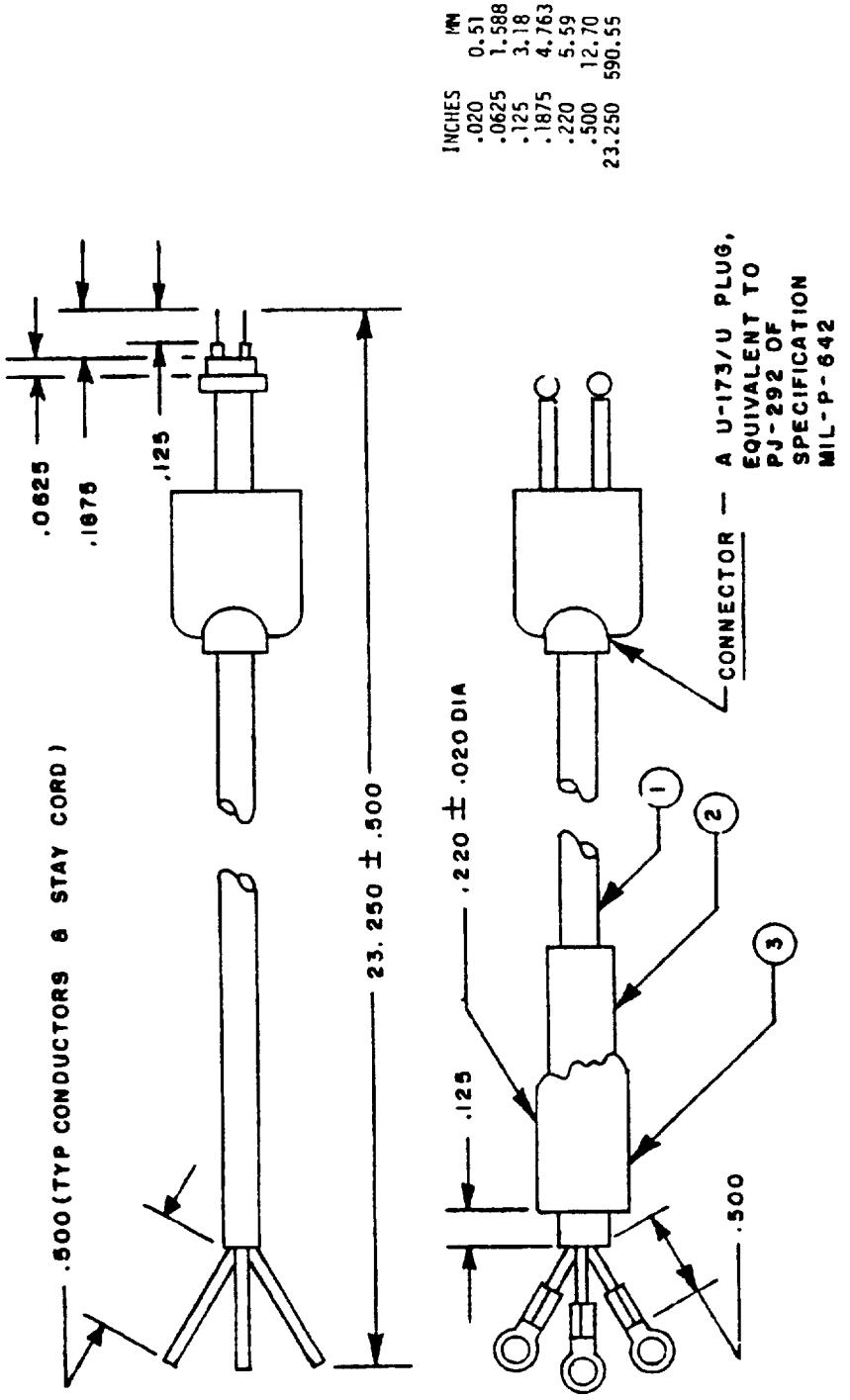


INCHES	MM
.002	0.05
.015	0.38
.030	0.76
.078	1.98
.125	3.18
.166	4.22
.187	4.75
.250	6.35
.312	7.92

NOTES:
1. Dimensions are in inches.
2. Material: Inner sleeve: Beryllium copper 25, per QQ-C-530.
   Outer sleeve: Half hard brass 22, per QQ-B-627.
3. Finish: (Outer sleeve only)
   Tin, .0002" minimum, per MIL-T-10727, type 1.
4. Unless otherwise specified, tolerances are ±.015 (0.38 mm).

FIGURE 2. Connector.

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INCHES	MM
.020	0.51
.0625	1.588
.125	3.18
.1875	4.763
.220	5.59
.500	12.70
23.250	590.55

NOTES
1. Heat shrink item 2 on item 1, then heat shrink item 3 on item 2. Caution: Do not heat shrink both item 2 and 3 at the same time.
2. Attach staycord terminal as close as possible to end of outer jacket.
3. Unless otherwise specified, tolerances are ±.015 (38 mm).

FIGURE 3. Connectors.

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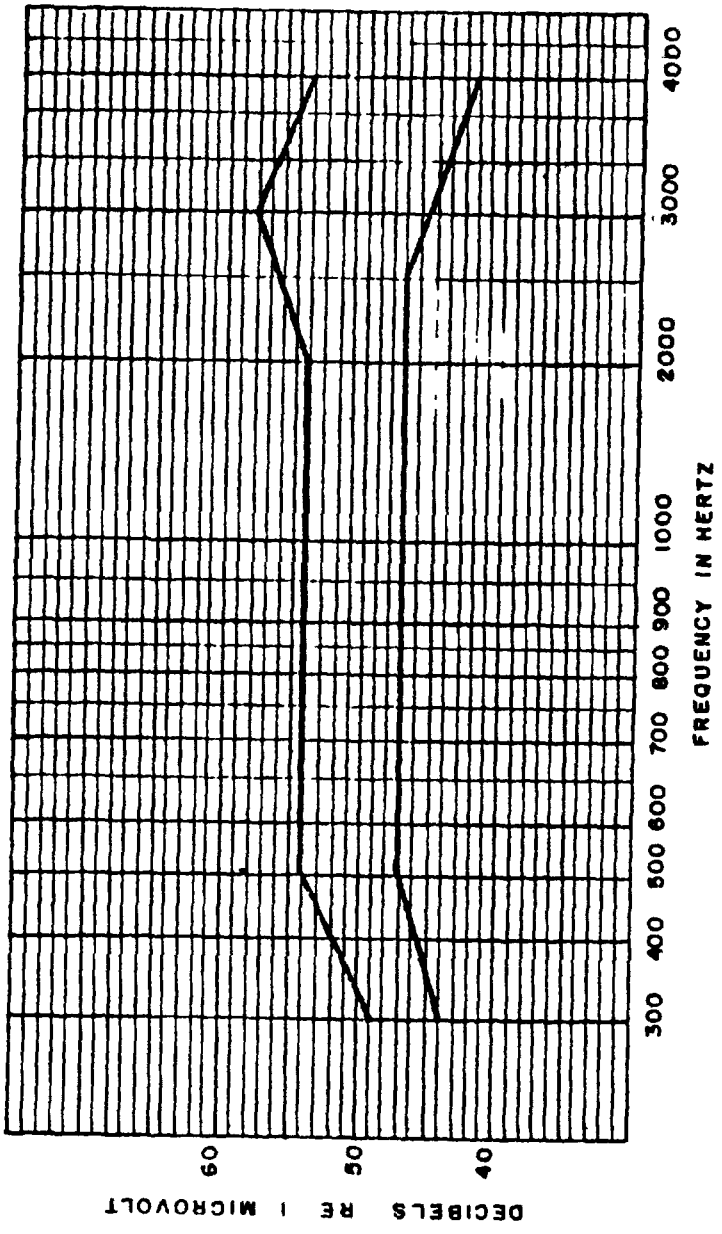


FIGURE 4. Frequency response.

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REQUIREMENTS:

Color: Semigloss black, 27038 in accordance with FED-STD-595.

Component parts:

Microphone element. See figures 1 and 2.

Bracket assembly. See figure 1.

Cord assembly: See figures 1 and 3.

Impedance: 4-6 ohms.

Resistive load: 5 ohms.

Performance:

Sensitivity at ground level. 51.5 ±3 dB above 1 microvolt with input of 115 dB SPL above 20 micropascals (0.002 dyne/cm^2) at 1 kHz.

Sensitivity at 10,000 feet: Not more than 3 dB below sensitivity at ground level.

Frequency response: The frequency response of the microphone, compared to the response of 51.5 ±3 dB at 1,000 Hz shall be within the limits shown on figure 4. However, the difference between any two successive readings 100 Hz apart up to 1,000 Hz and 250 Hz apart above 1,000 Hz shall not exceed 2.5 dB under all standard and service conditions. The response shall fall off at a rate of not less than 3 dB per octave below 300 Hz and 6 dB per octave above 4,000 Hz.

Distortion: The harmonic distortion in the microphone shall not exceed 4 percent under standard conditions or 5 percent after being subjected to service conditions.

Pressure equalization: The response of the microphone at 1,000 Hz shall not vary by more than 4 dB from the reference established at 15,000 feet while the microphone is subjected to a pressure increase corresponding to a descent from an altitude of 15,000 feet (16.7 inches of mercury) to sea level (30.0 inches of mercury) at a rate of 5,000 feet per minute.

Stray magnetic field: The stray magnetic field of the microphone shall not cause more than a 3 degree deflection when placed within 8 inches of a magnetic compass.

Weight: The weight of the microphone shall not exceed 1.5 ounces.

Interchangeability: The equipment shall meet the interchangeability requirements as defined in specification MIL-E-5400.

The microphone assembly shall be tested in accordance with the tests listed in table I.

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Ⓐ TABLE I. Parameter applicability.

Inspection	Requirements paragraph	Test paragraph	Qualification tests	Group "A" tests	Group "B" tests	Group "C" tests
Group I						
Visual and mechanical inspection	3.1,3.4,3.5	4.7.1	X	X		
Sensitivity of ground level	3.5.1	4.7.2.2	X	X		
Sensitivity at altitude	3.5.2	4.7.2.3	X			
Frequency response at ground level	3.5.3	4.7.2.4	X	X		
Frequency response at altitude	3.5.4	4.7.2.5				
Impedance	3.5.5	4.7.3	X	X		
Noise cancellation characteristics	3.5.7.1	3.7.5.1				
Effect of external magnetic field	3.5.8	4.7.6				
Stray magnetic field	3.5.9	4.7.7	X		X	
Linearity	3.5.10	4.7.8	X		X	
Talk out	3.5.11	4.7.9				
Dielectric withstanding voltage	3.5.12	4.7.10				
Signal to noise	3.5.7	4.7.5	X			
Distortion	3.5.6	4.7.4	X			
Interchangeability	3.5.13	4.7.11				
Group II						
Thermal shock	3.5.14	4.7.12	X			X
Humidity	3.5.15	4.7.13	X			X
Drop	3.5.16	4.7.14	X			X
Pressure equalization	3.5.17	4.7.15	X			X
Explosive decompression	3.5.18	4.7.16	NA			
Salt fog	3.5.19	4.7.17	X			X
Group III						
Vibration	3.5.20	4.7.18	X			X
Bounce	3.5.21	4.7.19	NA			
Altitude	3.5.22	4.7.20	X			X
Moisture barrier	3.5.23	4.7.21	X			X
Immersion	3.5.24	4.7.22	NA			
Group IV						
Fungus	3.5.25	4.7.23	NA			
Group V						
Gun blast	3.5.26	4.7.24	NA			

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Navy - AS
Air Force - 99
DLA - ES

User activities:
Army - AR, AT, AV, ME
Navy - OS

Agent:
DLA - ES

Preparing activity:
Air Force - 95

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